Ashrafhusen Raj

Data Engineer

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Summary:

- Data Engineer with 7+ years of experience designing and delivering scalable batch, streaming, and Lakehouse data platforms across FinTech, Healthcare and Industrial domains, with strong expertise in analytics, enterprise data quality, governance, and complex data flow analysis in highly regulated environments.
- Strong hands-on expertise in SQL, Python, Apache Spark (PySpark, Scala), Kafka, Flink, Airflow and dbt, building resilient ETL/ELT pipelines with integrated data quality, lineage, reconciliation, and observability.
- Strong analytics-oriented engineering background with experience analyzing large-scale enterprise datasets, tracing complex data flows, and improving data quality across upstream and downstream systems.
- Extensive experience across cloud and hybrid data platforms, including AWS (S3, Glue, Redshift, EMR, Lambda, EKS), Azure (Databricks, Synapse, Data Factory, ADLS, Event Hub), GCP (Big Query, GCS, Dataflow), dbt and Snowflake delivering cost-optimized, governed Lakehouse architecture.
- Proven expertise in Delta Lake, Apache Hudi, and Apache Iceberg, enabling CDC pipelines, schema evolution, partitioned ingestion, and time-travel analytics using Databricks and Unity Catalog.
- Skilled in data modeling and warehouse design (star and snowflake schemas) across Snowflake, Redshift, Synapse, SQL Server, and PostgreSQL, with afocus on performance tuning and scalability.
- Enabled BI, analytics, and business teams by designing trusted, analytics-ready datasets, semantic layers, and governed reporting structures consumed by Tableau and Power BI dashboards.
- Experienced in root cause analysis, data reconciliation, and behavioral/process-level issue identification to improve data accuracy, governance, and reporting reliability.
- Applied CI/CD and Infrastructure-as-Code practices to production-grade data platforms using Git/GitHub, Terraform, and Kubernetes, supporting automated deployment, serverless orchestration, event-driven architecture, versioning, and rollback of data pipelines.

Technical Skills:

Languages	SQL, Python, Java, Scala
Databases, Warehouses & Storage	**Relational & NoSQL:** PostgreSQL, MySQL, MongoDB **Cloud Data Warehouses:** Snowflake, Amazon Redshift, Azure Synapse, Big Query **Data Lake Storage:** AWS S3, ADLSGen2, Azure Blob Storage, GCS **Lakehouse Formats:** Delta Lake, Apache Hudi, Apache Iceberg
Big Data & Streaming	Apache Spark (PySpark, Spark SQL), Apache Kafka, Apache Flink, Hadoop, AWS EMR, Azure Event Hub
ETL, ELT & Orchestration	Apache Airflow, AWS Glue, Azure Data Factory, dbt, Matillion, Informatica
Cloud& Data Platforms:	**AWS** • S3, Glue, Athena, Redshift, RDS • AWS Lambda, Step Functions, Event Bridge, CloudWatch • Lake Formation, EMR, EKS, ECS, IAM **Azure** • Databricks, Synapse, Data Factory • ADLS • Event Hub, Azure Stream Analytics **GCP** • Big Query, GCS, Dataflow
Data Modeling & Architecture:	Star Schema, Snowflake Schema, Dimensional Modeling, Data Lake Architecture, Event-Driven Architecture
Version Control & SDLC	Git, GitHub, GitLab, Agile/Scrum
DevOps, Automation & IaC	CI/CD Pipelines (GitHub Actions, GitLab CI, Azure DevOps), Terraform, Kubernetes
Data Visualization & Analytics:	Power BI, Tableau, Amazon Quick Sight, Pandas, NumPy, Matplotlib, Seaborn
Data Quality, Governance & Security	Great Expectations, Unity Catalog, Data Lineage & Metadata Management, Schema Registry, Data Validation, AWS IAM, Azure Entra ID, **Compliance**: PII, HIPAA, SOC 2, PCIDSS, GDPR, SOX, IAM & RBAC, Data Encryption (at rest & in transit), Data Masking & Tokenization, Audit Logging & Data Lineage, Root Cause Analysis, Data Reconciliation, Data Profiling
Enterprise Applications & ERP/CRM Integration	Salesforce (Sales Cloud, Service Cloud, REST/SOAP APIs, Bulk API), SAP ERP (ECC / S/4HANA), SAP CRM, SAP HANA

Professional Experience:

United Health Group **Remote, USA**
Senior Data Engineer (Consultant) **Feb 2026 – Current (Aug)**

Responsibilities**:**

- Designed and implemented end-to-end ETL/ELT pipelines in Snowflake and Azure using complex SQL, stored procedures, and Python to ingest, transform, and consolidate data from on-prem and cloud sources for healthcare analytics and reporting.
- Worked with core healthcare data domains including claims, eligibility, provider, and member enrollment data, transforming raw feeds into analytics-ready datasets in Snowflake and Delta Lake.
- Standardized and harmonized data from multiple payers and providers using healthcare data standards such as HL7 v2, FHIR, and CCDA, ensuring consistent semantics and interoperability across downstream analytics and reporting.
- Orchestrated batch and near-real-time workflows in Azure Data Factory and Airflow on Kubernetes, implementing parameterized, metadata-driven designs for scalable ingestion and transformation of structured and semi-structured data (fixed-width, delimited, JSON, XML, HL7 messages).
- Implemented HIPAA-aligned data handling practices in pipelines, including encryption**,** minimum necessary access, audit logging, and PHI masking/de-identification in Snowflake and Databricks.
- Developed and optimized Databricks (PySpark) jobs and notebooks for large-scale data processing into the Magnus Lakehouse environment, following Medallion Architecture patterns (bronze/silver/gold layers).
- Performed data modeling (star schema, dimensional models) and query tuning in Snowflake to optimize performance and cost for analytics, risk adjustment, and care gap use cases.
- Leveraged AI-assisted development tools (e.g., GitHub Copilot, Databricks AI Assistant) to accelerate PySpark and SQL pipeline development, testing, and documentation while maintaining enterprise coding and security standards.

Lexmark Inc. **Lexington, KY**
Data Engineer **Mar 2023 – Jan 2026**

Responsibilities:

- Designed and enhanced cloud-based data platforms across Azure and AWS ecosystems, designing scalable ETL/ELT pipelines using Azure Data Factory, Databricks, Synapse Analytics, S3, Lambda, and Redshift to support high-performance analytics and governed enterprise data workflows.
- Managed AWS cloud infrastructure (EC2, S3, SQS, SNS) and maintained Redshift data warehouse, optimizing performance, reliability, and scalability of data pipelines and storage systems.
- Engineered batch and real-time event-driven pipelines using **PySpark and Spark Structured Streaming**, integrating **Apache Kafka** for millions of event records per day, supporting behavioral analytics and operational monitoring.
- Implemented enterprise data governance using Databricks Unity Catalog, enforcing access controls, column-level masking, and GDPR/CCPA compliance across 2000+ data assets to improve governance adoption and audit efficiency.
- Built Snowflake + dbt ELT pipelines, transforming structured and semi-structured data (JSON, CSV, Parquet) from ADLS and REST APIs into analytics-ready, high-quality datasets.
- Designed and developed Azure Data Factory pipelines to migrate and synchronize Salesforce data into ADLS and Synapse Analytics; implemented data validation, transformation logic, and audit tracking to ensure data integrity, lineage, and compliance.
- Designed and developed data models (CDM, LDM, PDM) and versioning strategies, enabling 100% historical data traceability and improving query performance for analytical workloads by 30% through optimized schema design.
- Optimized ETL and query performance with partitioning, indexing, and Spark SQL tuning, achieving 40% faster pipeline processing times. Migrated **legacy on-prem workflows to Databricks Delta**, ensuring ACID compliance, operational reliability, and adherence to **data contracts and governance policies**.
- Supported migration of large-scale data pipelines from AWS to Azure enhancing ETL workflows for cloud-native performance and scalability.
- Supported SAP ERP (CRM/SD) implementation by designing data extraction and validation pipelines from legacy Salesforce and operational systems into SAP staging environments; ensured reconciliation, schema alignment, and business rule validation during migration phases.
- Designed and orchestrated scalable ELT pipelines using **Matillion** and Snowflake, integrating data from APIs, ADLS, and relational sources; leveraged push-down transformations to optimize warehouse performance, reducing data processing latency by 30%.
- Performed enterprise data flow analysis and root cause investigations across integrated ERP, CRM, and analytics platforms to identify upstream/downstream impacts, improving data consistency and reporting accuracy.
- Monitored pipelines and ensured observability with **Airflow and Splunk**, proactively identifying bottlenecks, reducing incident resolution time, and performing **on-call data engineer responsibilities**.

Bank of America **Chicago, IL**
Data Engineer **May 2021 –Feb2023**

Responsibilities:

- Designed and deployed high-throughput event-driven ETL pipelines using Apache Spark on AWS EMR, processing terabyte-scale datasets for operational and financial reporting, optimized Spark job execution through dynamic partitioning and columnar storage.
- Migrated legacy systems to Bank of America's private cloud data platform, orchestrating end-to-end ETL workflows and ensuring performance and consistency through rigorous post-migration validation with zero data loss.
- Designed and orchestrated scalable ETL pipelines on AWS using AWS Glue, Lambda, and Spark to process legacy and API-driven data sources, enabling event-driven integrations and curated datasets for analytics and operational reporting.
- Designed and developed event-driven streaming pipelines using Kafka (MSK) and Spark Structured Streaming, implementing checkpointing and idempotent writes to Delta Lake for reliable, fault-tolerant processing and real-time data integration.
- Ingested and transformed structured and semi-structured data from Teradata, Oracle, mainframe, and SaaS systems into cloud data lakes using JDBC connectors and API integrations, enabling enterprise analytics and downstream reporting.
- Converted raw ingestion into optimized formats (Parquet, Avro, JSON) for schema evolution, efficient storage, and compatibility with Spark and Redshift; managed data versioning and automated schema registry for Kafka topics ensuring backwards compatibility.
- Designed and implemented Databricks ELT workflows in Spark and Scalafor large scale transformations and downstream reporting.
- Implemented a three-zoneData Lake structure (raw, staging, curated) using Delta Lake for transaction-safe updates and schema enforcement.
- Built and scheduledcomplex multi-stage datapipelines using Apache Airflow, including custom Python-based tasks for validation, alerting, and anomaly detection.
- Conducted root cause analysis on enterprise data discrepancies by tracing behavioral, process, system, and integration-level drivers; implemented automated reconciliation and Spark validation checks to improve data quality and operational reporting reliability.
- Published data lineage and metadata to AWS Glue Catalog and enterprise governance tools, enforced schema validation and data classification policies in downstream pipeline.
- Enhanced ETL performanceby parallelizing workflows, indexing, andoptimizing SQL transformations, achieving up to 30% reduction in data processing times.
- Developed optimized cloud data warehouse architectures with optimized schema design, columnar compression, and distribution keys; implemented incremental ingestion and automated backfilling for historical data loading.
- Developed governance and security practices including PII data masking, encryption at rest (AWS KMS) and in transit (TLS), and IAM-based access controls; ensured PCI-DSS compliance across sensitive financial data.

Chase Bank (JPMC) **Chicago, IL**
Data Engineer / ETL Developer **Jul 2019 –Apr 2021**

Responsibilities:

- Migratedon-prem Oracle databases to Amazon Redshift, optimizing distribution/sort keys; achieved significant improvements in query performance and infrastructure costs.
- Built end-to-end ETL pipelines on Databricks using Spark, Scala, Kafka, Flink, and AWS Glue to ingest, process, and store structured and unstructured data on a scale.
- Designed datasolutions with DatabricksDelta Lake, data warehouses, and data marts to support analytics and data science workloads.
- Developed dimensional data models (star and snowflake schemas) to improve warehouse performance and reduce query latency.
- Designed and implemented data quality frameworks using Informatica Data Quality (IDQ), including data profiling, rule development, cleansing workflows, and monitoring dashboards to ensure data accuracy, consistency, and compliance across enterprise systems.
- ImplementedCI/CDdeployment pipelineswith Terraform and Airflow for automated orchestration and infrastructure provisioning.
- Integrated Flink pipelines with Schema Registryto manage schema evolution and maintain backward compatibility across producers and consumers.
- Designed and implemented Apache Flink streaming applications for high-volume, low-latency event processing, integrating with Kafka topics for real-time analytics and operational monitoring.
- Built SQL-driven analytical reporting solutions by translating business and operational requirements into optimized queries, curated datasets, and visualization layers for enterprise stakeholders.
- Designedingestion frameworks leveraging AWSservices including EC2, RDS, S3, Athena, Glue, Lambda, EMR, Kinesis, and SQS for scalable processing.
- Partnered with ML engineers to optimize data pipelines for model training; reduced feature engineering time by 40% through Spark optimizations.
- Created and optimized data models and storage solutions across SQL, NoSQL, key-value stores, and data lakes to ensure data quality and accessibility.
- Optimized query performance by designingefficient indexes andviews; createdreusable stored procedures and UDFs to reduce code duplication across ETL workflows.
- Conducted performance tuning for ETL jobs anddatabase queries to improvethroughput and resource utilization.

Education:

BachelorofEngineering in InformationTechnology

(Gujarat Technological University)

CERTIFICATION:

Oracle Cloud Infrastructure - Data Science Professional

AWS certified Data Engineer- Associate